



EAST TEXAS A&M
— UNIVERSITY —

CSCI 380.61E
WEB PROGRAMMING
AND INTERFACE DESIGN

OVERVIEW

- What do we mean by Productive Web Programming?
- **Good** and **bad** practices
- Structure of the Web
- Working as a **team**
- Version control (GitHub)
- Testing and validation
- **Open source** practices
- Questions and Discussion

REVISION QUESTION

Is the following HTML correctly formatted?

```
<b><font face="arial narrow"><i>  
I am bold, italicized, arial narrow text :)  
</i></b></font>
```

ANSWER

NO - The HTML tags are improperly **nested**.

Incorrect

```
<b><font face="arial narrow"><i>  
I am bold, italicized, arial narrow text :)  
</i></b></font>
```

Correct

```
<font face="arial narrow"><b><i>  
I am bold, italicized, arial narrow text :)  
</i></b></font>
```

WHAT DO WE MEAN BY PRODUCTIVE WEB PROGRAMMING?

WHAT DO WE MEAN BY PRODUCTIVE WEB PROGRAMMING?

- Structure
- Reusability
- Robustness
- Scalability
- ..

WHY IS STRUCTURE IMPORTANT?

- So we understand how everything **fits together**
- Bigger picture
- Logical and **reliable**
- **Client-server** interaction
- What goes where and why (e.g., client or server)

Three-tier Architecture



Front-end Applications

JavaScript, HTML5, CSS3, XML

CLIENT

Web Applications

UI-Business integration

- PHP
- Python



Compiled Software Applications

Resolve business problems

- JAVA
- C++ - Python

Back-end Applications

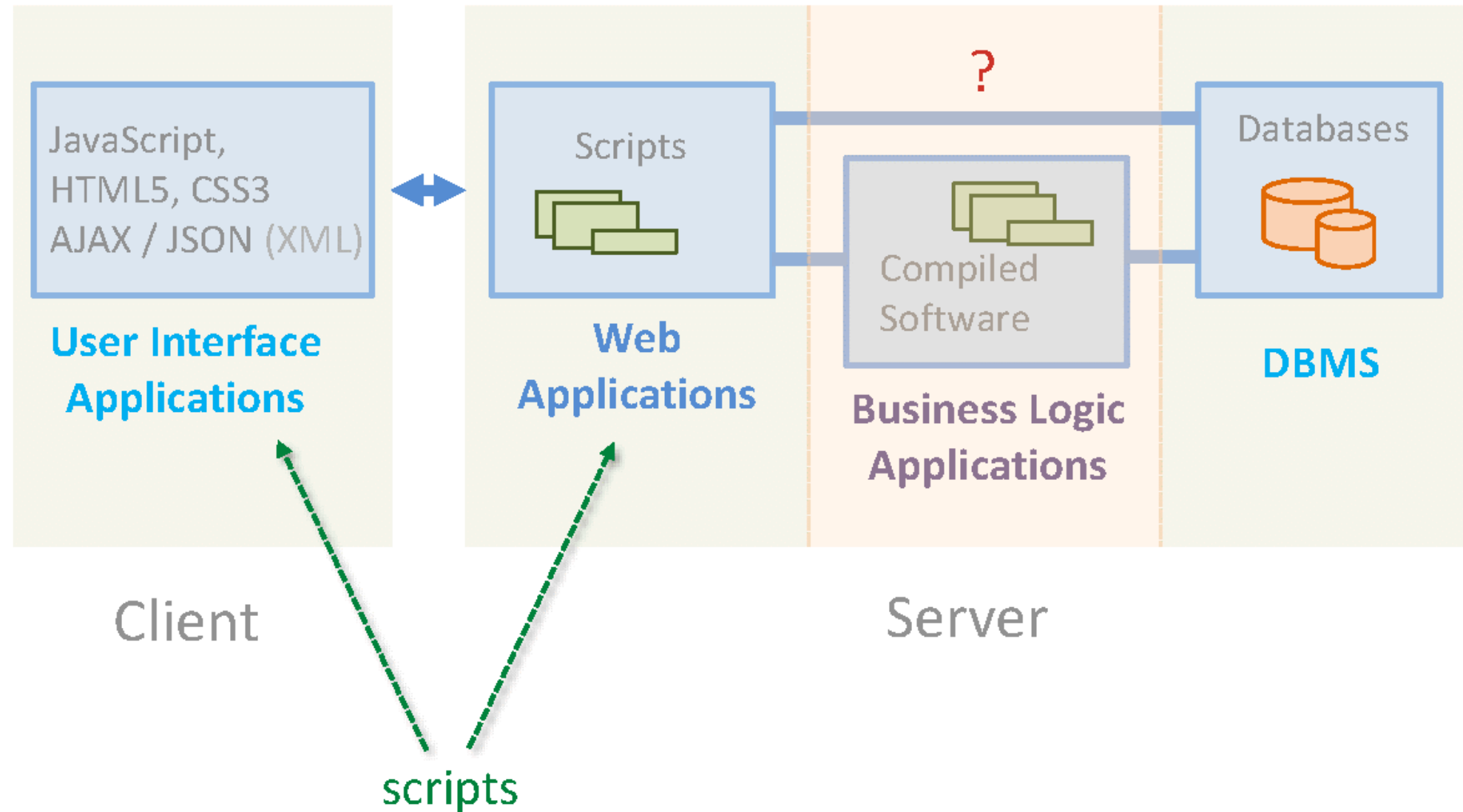


Database Applications

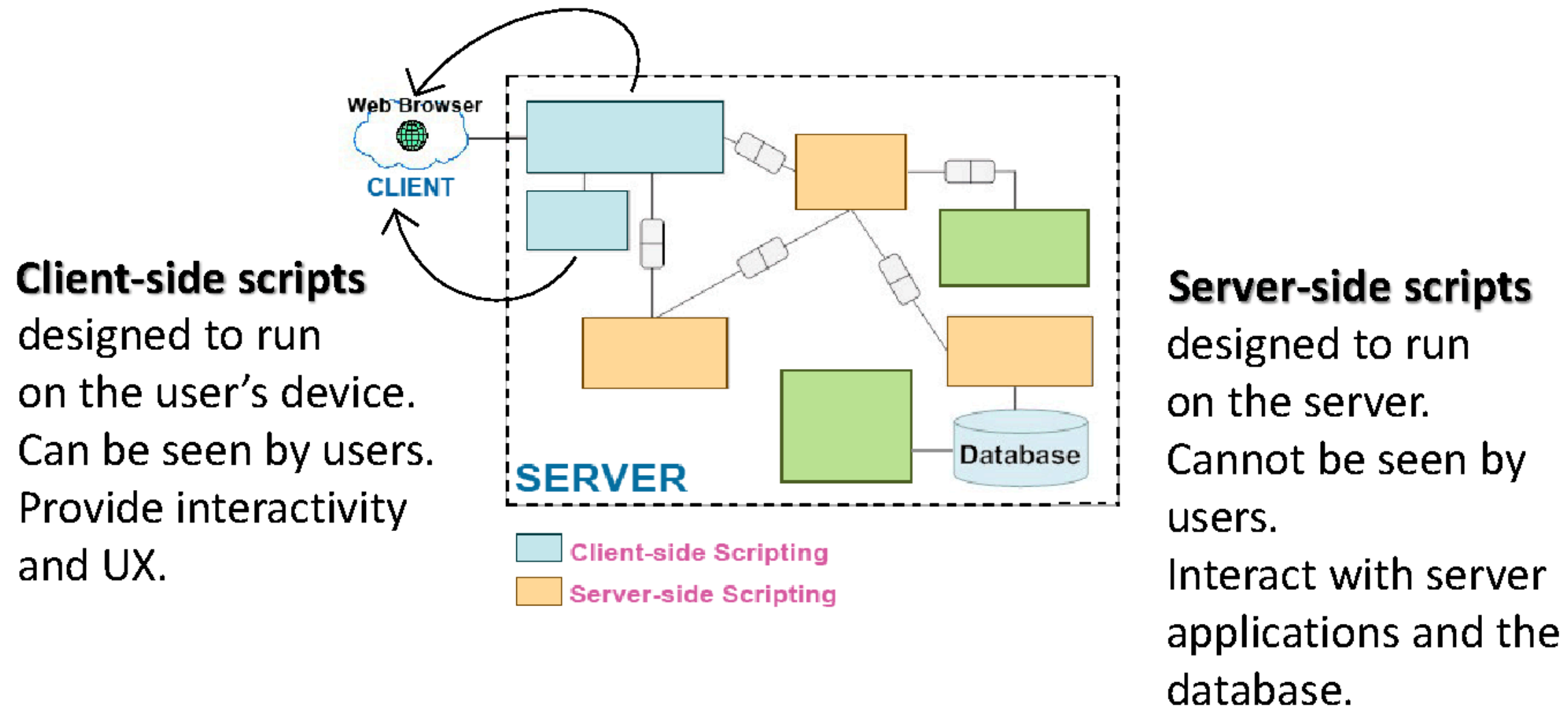
- Relational DBMS (MySQL)
- NoSQL DBMS (MongoDB)
- Data Warehouse (Drill)

SERVER

Web Client-Server Architecture



Client-side and Server-side Scripts



Scripting Languages



A scripting language was designed to **glue** together [compiled] applications (e.g. business applications)

But now they are more than glue:

- Database Interfaces
- Built-in Functions
- Libraries
- ...

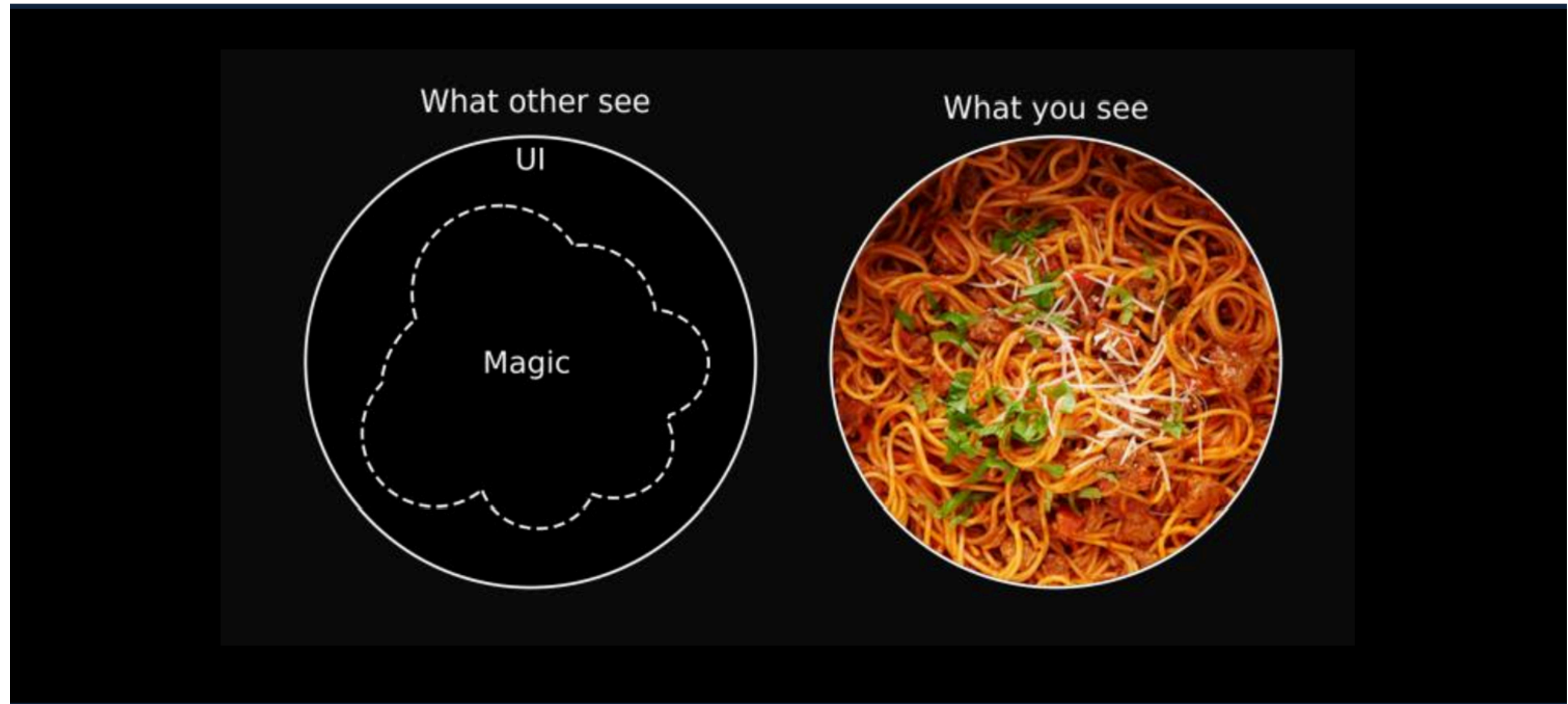
Web Application = Scripting



Web Applications $\neq$ Compiled Software Applications
with Resources-Management capabilities

In Web Scripting

it's easier to make mistakes & get bad habits



WEB SCRIPTING GOOD PRACTICES

Web Scripting Best Practices 1



1. Think Maintenance

It's harder to read code than to write it

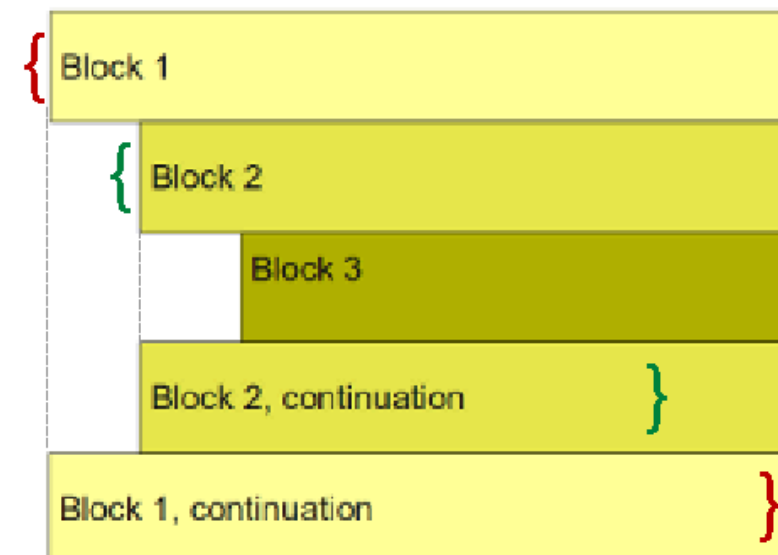
"Code is like humour. When you have to explain it, it's bad."
(Cory House)

Web Scripting Best Practices 1

1. Think Maintenance

HOW?

Always
INDENT
your
script
in blocks



Web Programming Best Practices 1

1. Think Maintenance

HOW?

comments are to explain **WHY**, not what is done



```
/**
 * Simple HelloButton() method.
 * @version 1.0
 * @author john doe <doe.j@example.com>
 */
HelloButton()
{
    JButton hello = new JButton( "Hello, wor
    hello.addActionListener( new HelloBtnList

    // use the JFrame type until support for t
    // new component is finished
    JFrame frame = new JFrame( "Hello Button"
    Container pane = frame.getContentPane();
    pane.add( hello );
    frame.pack();
    frame.show();
}
```

Web Scripting Best Practices 1

1. Think Maintenance

HOW?

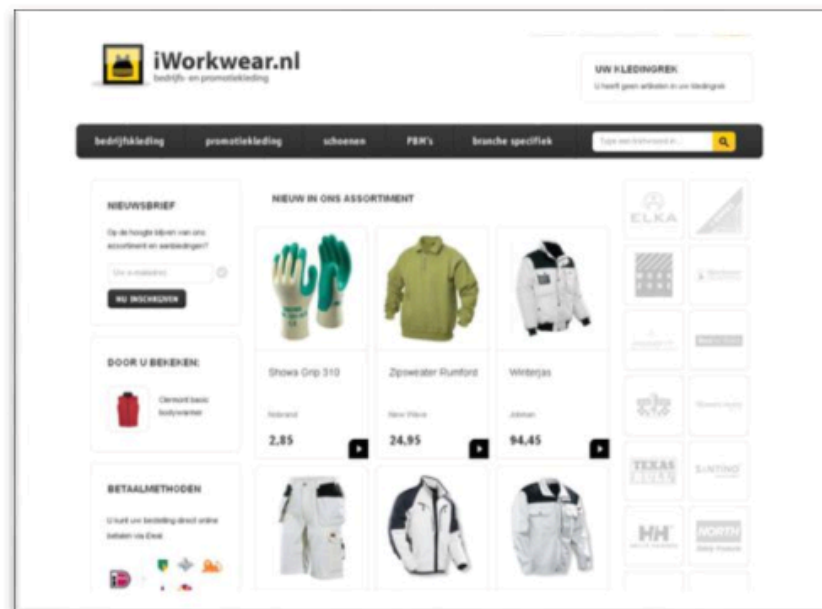
Organise your files and folders in a systematic way:

- ✓ Your application's file system should reflect your 3-tier architecture
- ✓ Scripts associated with a functionality should be in the same folder
- ✓ Use meaningful file/dir names (**what**/why/how)

Web Scripting Best Practices 1

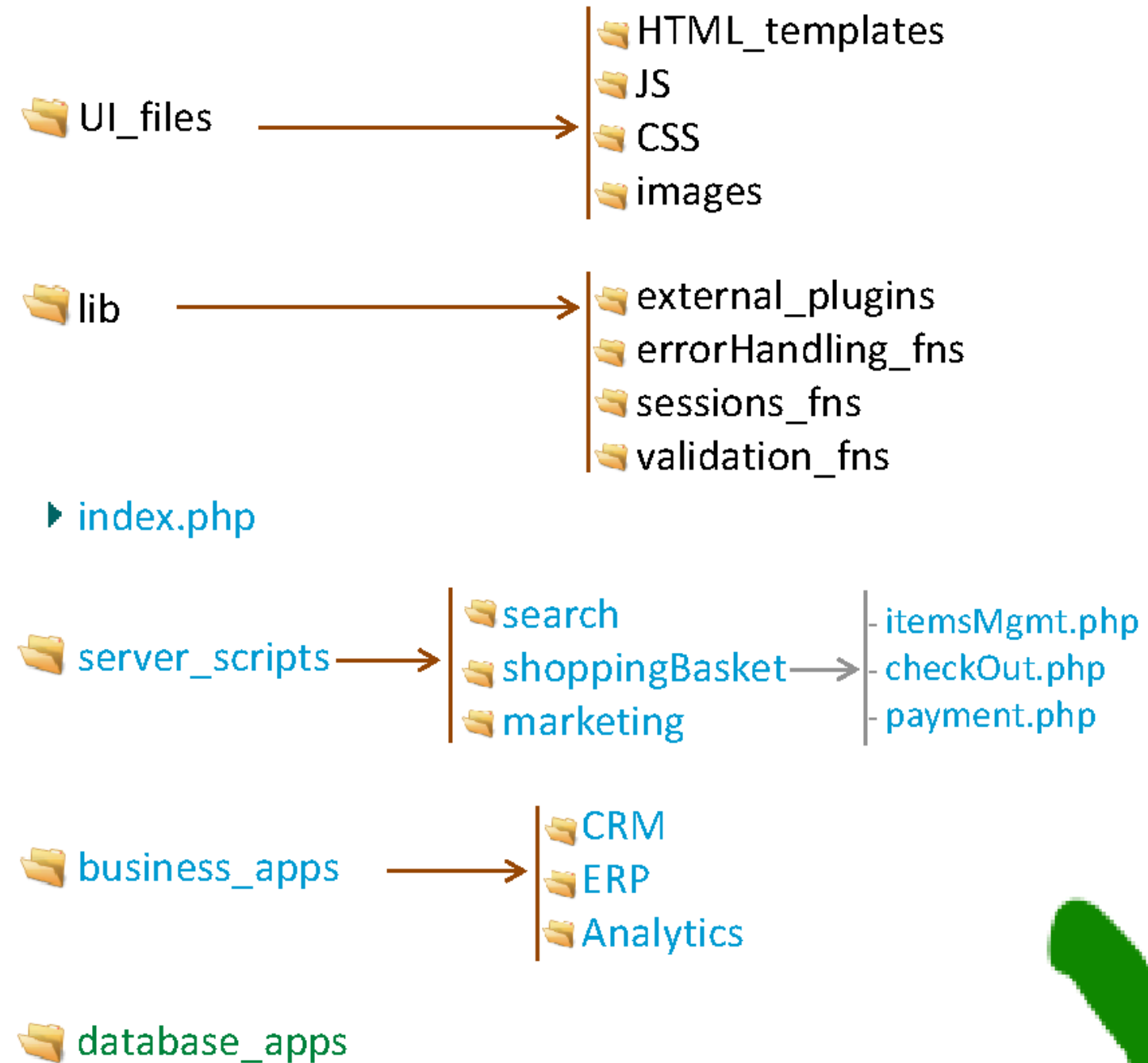
1. Think Maintenance

HOW?



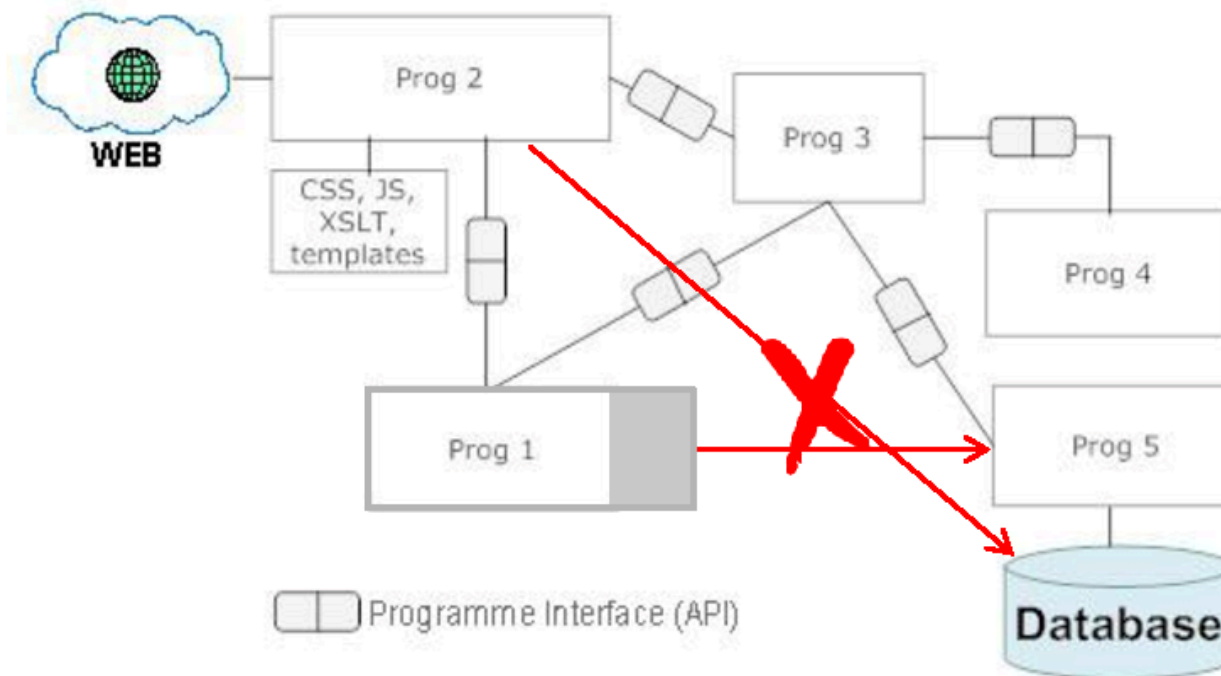
- index.php (loose file)
- images_and_templates
- webPages.php
- shopping.php
- miscellaneous applications

Web Scripting Best Practices 1



Web Scripting Best Practices 2

2. Write robust and **small** scripts connected by clear interfaces



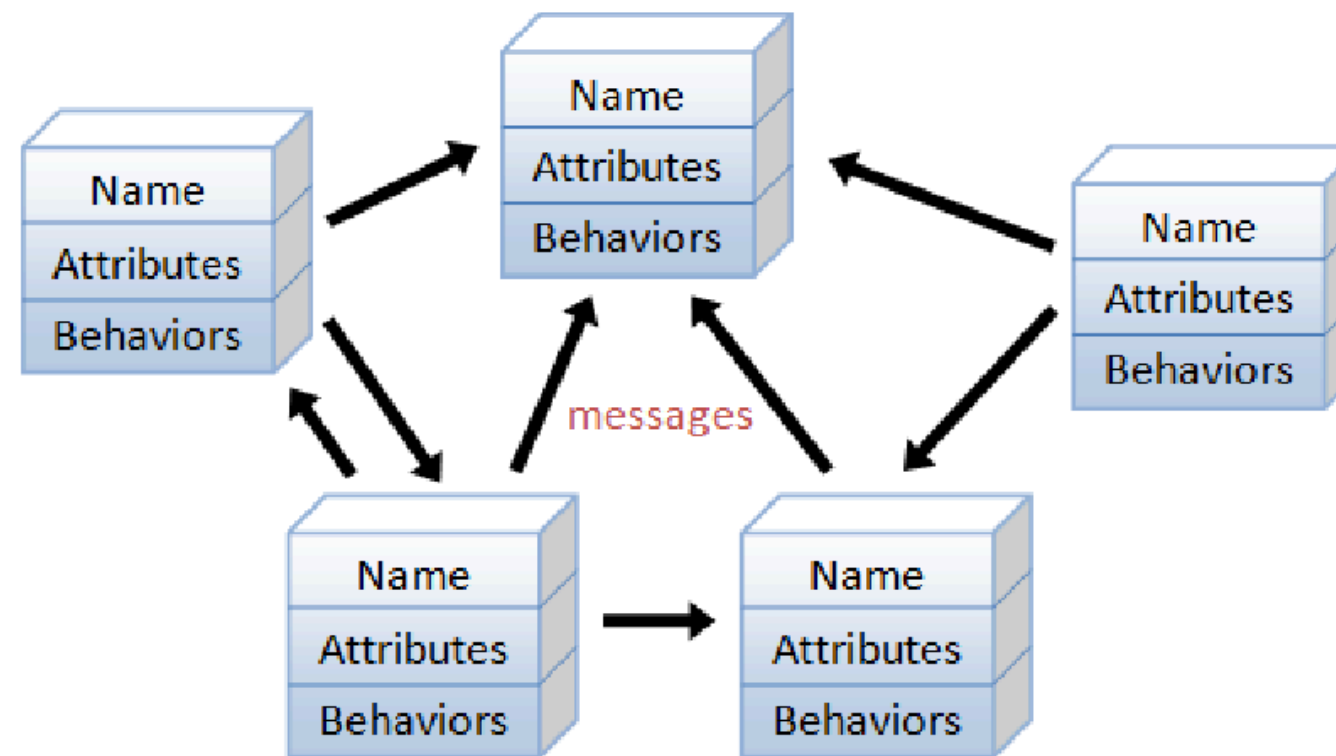
HOW?

Resist the temptation to directly connect UI scripts with Database queries

Resist the temptation to bypass your programme interfaces

Web Scripting Best Practices 2

Write robust and **small** scripts connected by clear interfaces



HOW?

Use Object-Oriented Programming (OOP) tactics

Web Scripting Best Practices 2

Write robust and **small** scripts connected by clear interfaces

HOW?

- Do not use your own Global variables
- Use built-in Global variables (superglobals)
- Ideally you should have one file per functionality

PHP Superglobals

- `$GLOBALS`
- `$_SERVER`
- `$_GET`
- `$_POST`
- `$_FILES`
- `$_COOKIE`
- `$_SESSION`
- `$_REQUEST`
- `$_ENV`

Web Scripting Best Practices 3

3. A script should do one thing and do it well

HOW?

Don't overdo it!

Do not to add functionality until
you need it.



Implement only
one functionality
per programme
(Single Responsibility)



Resist the temptation to add more functionality to the original scripts

Web Scripting Best Practices 3

A script should do one thing and do it well



- ✓ Code Review
- ✓ Unit Testing
- ✓ Functional Testing
- ✓ Integration Testing

Web Scripting Best Practices 4

Open/Closed Principle

Code should be open for extension, but closed for modification.

Write code that doesn't have to be changed every time the requirements change.

HOW?

OOP:

```
class Animal {  
    public $hungry = 'hell yeah.';  
    function eat($food) {  
        $this->hungry = 'Thank you.';  
    }  
}
```

```
class Dog extends Animal {  
    function eat($food) {  
        if($food == 'cookie') {  
            $this->hungry = 'not so much.';  
        } else {  
            $this->hungry = 'barf!';  
        }  
    }  
}
```

Open/Closed Principle

Code should be open for extension, but closed for modification.



Web Scripting Best Practices 4

Open/Closed Principle

Code should be open for extension, but closed for modification.

Write code that doesn't have to be changed every time the requirements change.

HOW?

Procedural Programming:

- ✓ Focus in reusable modular functions
- ✓ Small functions containing a single algorithm and the data it needs.
- ✓ Avoid using global variables

Beware of **do that AND that**

WHY USE VERSION CONTROL?

- Which companies use version control?
- What is the most popular version control system?
- Why wouldn't you use version control?

WHY USE VERSION CONTROL?

- **Industry standard**
- Track progress, changes, bugs, ..
- Safety (backup)
- Collaborately working (team)
- Flexible

GIT AND GITHUB

- Git is a version control standard
- GitHub is the social code-hosting platform
- Free and the one of the most popular choices for web development
- See Nodejs, Angularjs, Reactjs (on GitHub)



WHO USES GIT?



RIGHT TOOL RIGHT JOB



CHECKLIST WEB PROGRAMMING

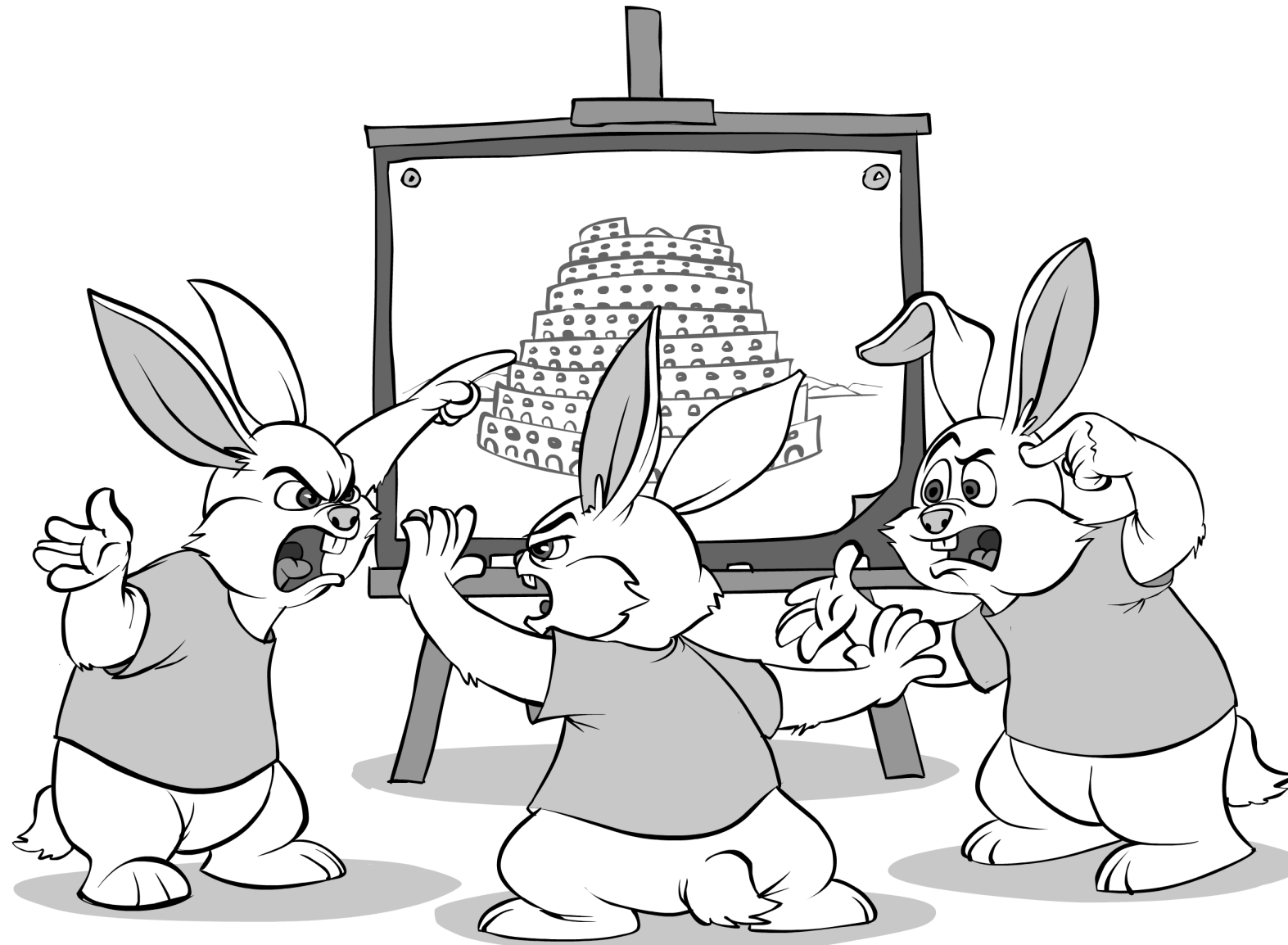


- Don't ignore good programming practices (just because it's **web**)

TEAM WORK

- Group work **essential**
- **Challenging** but also **important**
- Aspect of this module
- Learn and teach others
- **Skillset, abilities, ..**
- Appreciate **different perspectives**

TEAM WORK



Crucial tool for teamwork

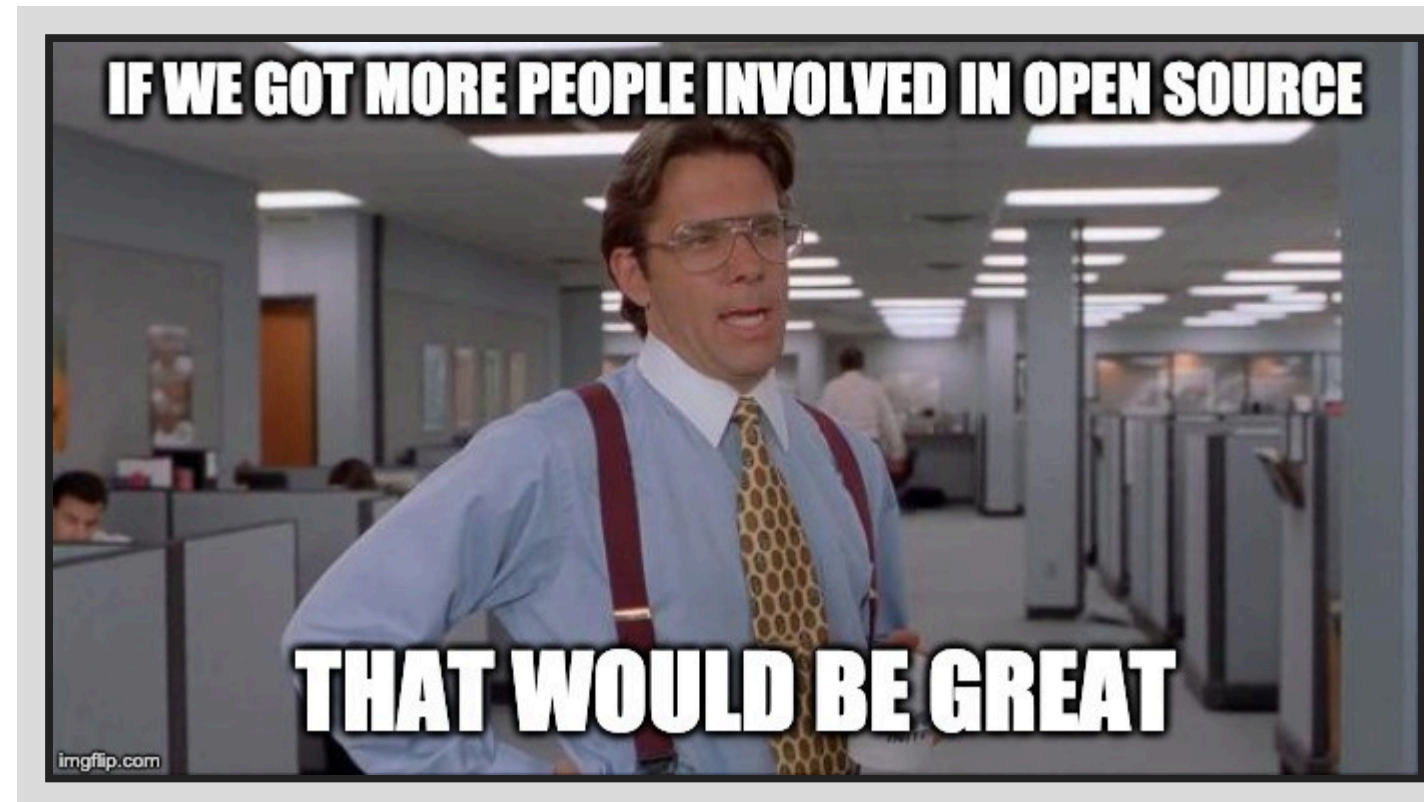
MANAGE/TRACK YOUR TEAM/FEATURES/TASKS



OPEN SOURCE PROJECTS

Are open source projects really that **important**?

OPEN SOURCE COMMUNITY



WHY OPEN SOURCE?

- Embrace open **innovation**
- Coding and **learning**
- Business card for skills (**portfolio**)
- **Community** support (bring ideas to life)

OPEN SOURCE GOOD PRACTICES

- Write useful **documents**
- **Standards** (e.g., header details for each file/copyright)
- Useful comments (not stories)
- **Help** others
- **Synergy**

ORGANISE ISSUES

- Track bugs
- Tasks
- Features
- Use available **tools** (e.g., GitHub)
- Plan

COMMUNITY

- Think **bigger** (other users)
- Reputation
- Utilizing users (testing/tips/..)
- **Showcase** with demos/videos
- Release **builds**
 - 0.1, 0.12, ...
 - features with each build

REVISION QUESTION

Can you freely use code/resources from a GitHub site if it has no license?

- no public declaration of license
- free to use/modify/sell or contribute to the software

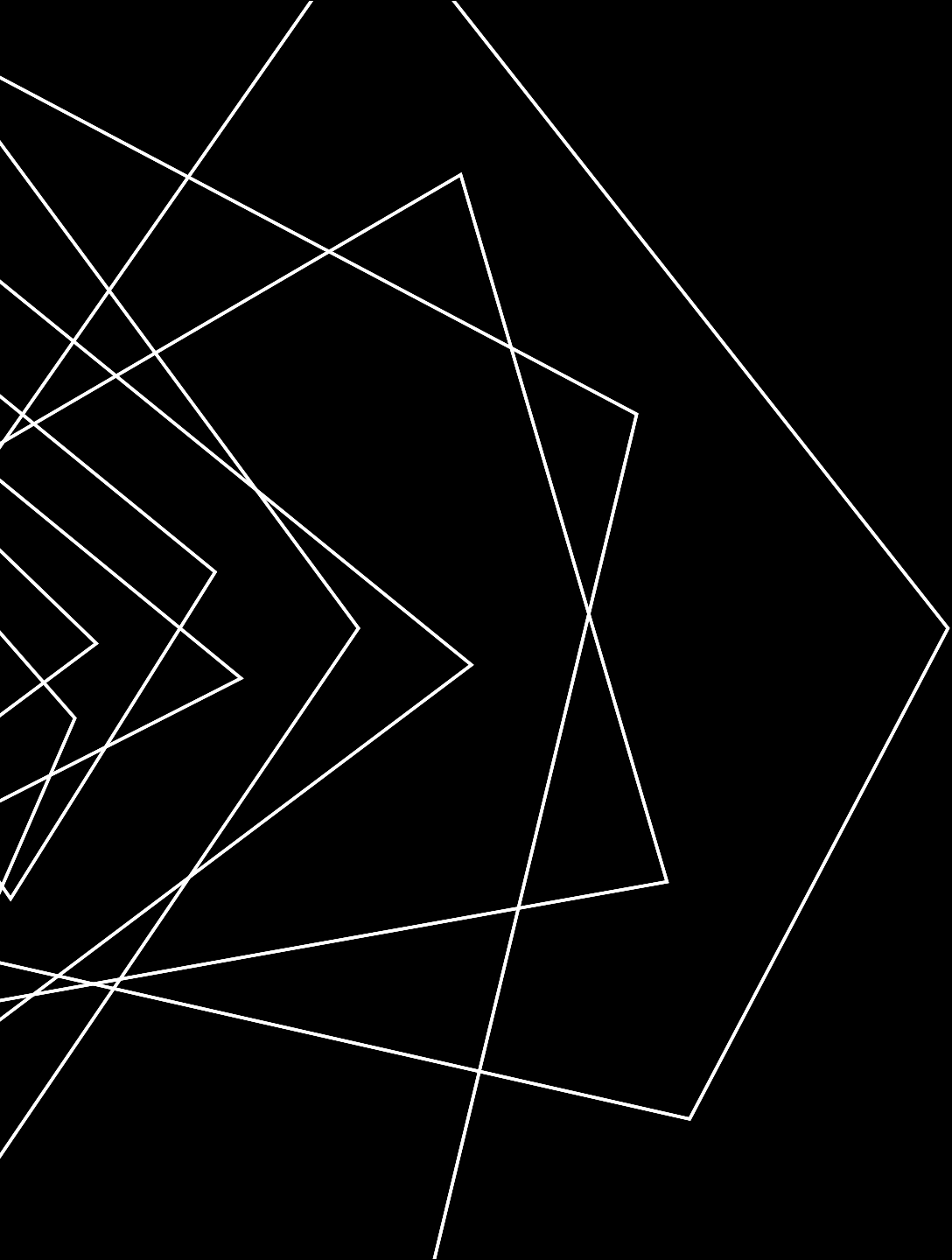
ANSWER

If a repository has no license, then **all rights are reserved** and it is **not** Open Source or Free. You **cannot** modify or redistribute this code without **explicit** permission from the copyright holder.

[Link Details](#)

SUMMARY

- Understand **Productive** Web Programming Concepts
- **Good and the bad** examples
- How to apply them in practice (e.g., **Coursework**)
- Real-world (standards, planning, features, bugs, ..)
- Open source (important and useful)



THANK YOU

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